THE LONG RUN, RECONCILED

10,000,000 rounds. *The interval holds.*

One hundred times the 100k sample. The same six decks, the same one-unit chart, and the same first 1,000,000 rounds.



NET RESULT

-41,722.50 units

W–L–P	OBSERVED EDGE	95% WIDTH
4,333,929–4,761,442–904,629	0.417%	0.143 pts

6 decks · S17 · 3:2 · Double any 2 · DAS · 3 resplits

SEED 8675309 · 75% PENETRATION · COMMIT 6e0fc28

THE PRECISION STORY

One edge. *Three error bars.*

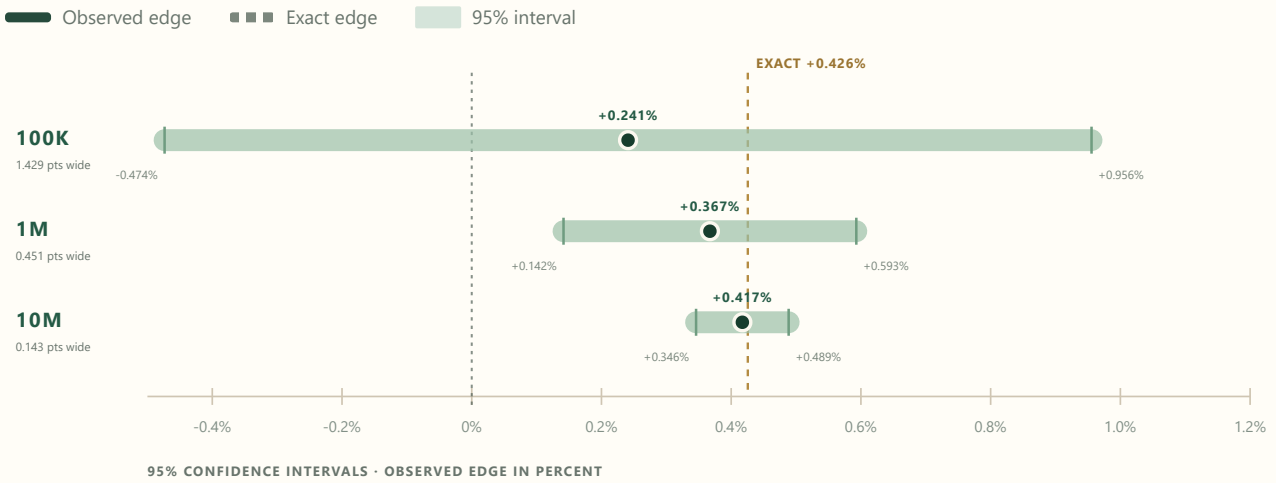
The same seed grows from 100,000 to 10,000,000 rounds. The 95% interval falls from **1.429 points wide** to **0.143** — a **10.0×** reduction. At 10M the observed edge is **0.417%**, and the interval covers the fresh-shoe **0.426%** reference.

OBSERVED EDGE
0.417%
10,000,000 rounds

EXACT EDGE
0.426%
composition-aware solver

95% INTERVAL
0.346%
to 0.489%

WIDTH
0.143 pts
10.01× narrower than 100k



95% CONFIDENCE INTERVALS · OBSERVED EDGE IN PERCENT

The 100k interval included break-even. The 1M interval did not. All three intervals cover the fresh-shoe reference, and the 10M interval is only 0.143 points wide.

Measure	100,000	1,000,000	10,000,000
Observed edge	0.241%	0.367%	0.417%
95% interval	-0.474% to 0.956%	0.142% to 0.593%	0.346% to 0.489%
Interval width	1.429 pts	0.451 pts	0.143 pts
Standard error	0.365 pts	0.115 pts	0.036 pts
Net result	-241.00 units	-3,672.00 units	-41,722.50 units
Error vs exact	+0.51σ	+0.51σ	+0.23σ

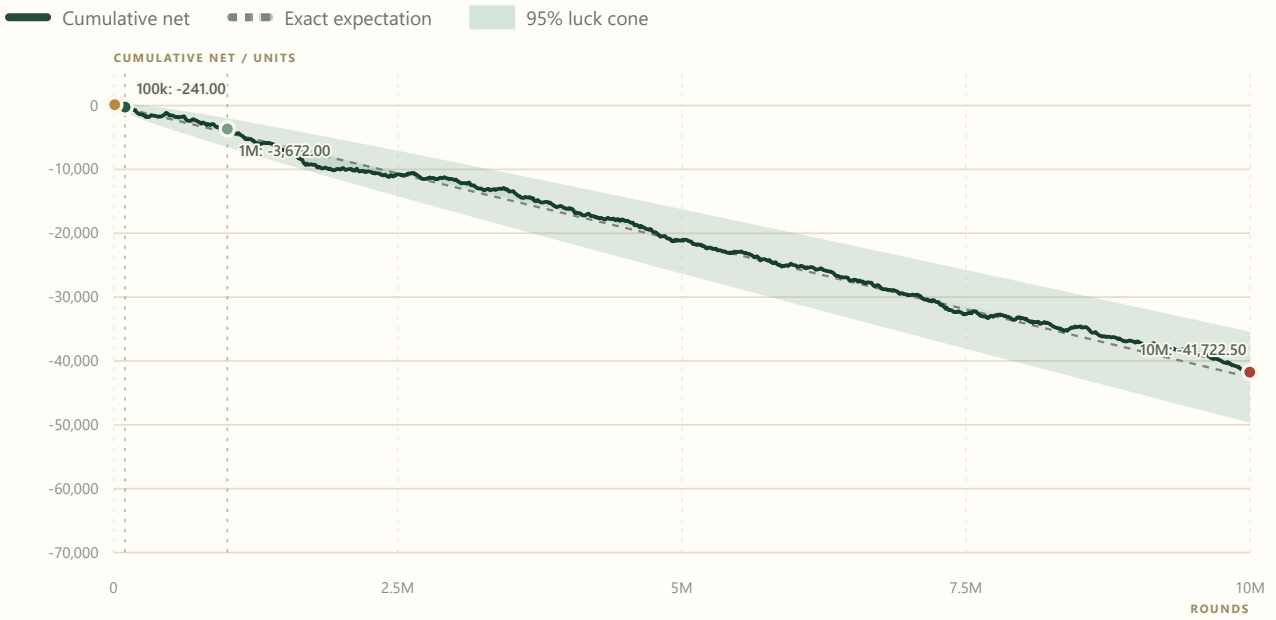
THE GAP WAS INSURANCE

The 10M interval is **0.346% to 0.489%**. The fresh-shoe exact edge is **0.426%**, inside it. The point estimate is **0.008 points below exact** — a 0.23σ difference. The previous 0.518% finish was not a model gap: it came from the post-hole insurance density. Correcting that estimator moves the same card sequence by 0.101 points.

TEN MILLION ROUNDS

The line stays *in the cone.*

Cumulative net over 10,000,000 rounds. The markers at 100,000 and 1,000,000 are the finishes of the smaller reports; the dashed line is the exact expectation and the cone is the 95% luck range. At 10M the path finishes inside it.



Expected path: 0 → -42,561.18. The 95% band at 10,000,000 rounds is roughly ±7,136.05 units. The finish is inside the band. Path sampled every 10,000 rounds.

PEAK	TROUGH	MAX DRAWDOWN	LONGEST LOSS STREAK
+133.00	-41,755.50	41,888.50	21
after round 8,844	after round 9,996,975	units, peak to trough	rounds in a row

At 100,000 the run was +184.61 units above exact expectation. At 1,000,000 it was +584.12 above. At 10,000,000 it was +838.68 above. All three finishes are inside the cone.

The 10M finish is 0.23 standard deviations from exact. The 95% band is ±7,136.05 units at the end. The finish is 838.68 units above expectation and inside the band.

WHAT ONE HUNDRED TIMES THE SAMPLE BOUGHT

The interval narrowed from 1.429 points at 100k to 0.451 at 1M and 0.143 at 10M. The observed edge moved 0.241% → 0.367% → 0.417%. After the insurance correction, the smallest interval lands on the exact path instead of missing it.

HOW THIS WAS PLAYED

More rounds. *A finer question.*

This is the same `PlayGame` table engine, rules and chart policy as the smaller reports. Per-decision logging is disabled so ten million rounds stay memory-flat.

Rules

Decks	6
Dealer	Stands on soft 17
Blackjack	3:2
Double	Any two · after split
Splits	Up to 4 hands · aces one card
Surrender	None
Dealer peek	US peek
Insurance	Offered; take when EV ≥ 0

Run

Seed	8675309
Rounds	10,000,000
Hands after splits	10,278,347
Bet	1 unit, flat
Penetration	75%
Strategy	Total-dependent chart
First 100k	-241.00 · first 1M -3,672.00
Code	6e0fc28

What changed

The 95% interval narrowed from 1.429 points at 100k to 0.451 at 1M and 0.143 at 10M. The observed edge moved 0.241% → 0.367% → 0.417%.

The first 100,000 and 1,000,000 rounds are not new simulations. They are the same nested sequences as the smaller reports.

What did not change

The rules, seed, cut card, chart and insurance threshold are identical. The exact fresh-shoe edge remains 0.426%. The 10M point estimate sits 0.008 points below it, well inside the interval.

Split EV still uses the documented independent-hand approximation. The chart is total-dependent, not composition-dependent.

What the correction changed

The earlier 0.518% result did not expose a model gap. Its insurance selector divided the post-hole ten count by the pre-hole unseen count, so a hidden ten made the shoe look poorer in tens than it was. At the 1/3 threshold that selected the wrong hands: 82,698 insurance bets won 26.9% of the time instead of the 34.9% their own densities implied.

Tracking visible cards instead moves the same 10M card sequence to **0.417%**, with insurance turning from a **0.080-point cost** into a **0.021-point gain** for the player. The exact 0.426% reference is inside the 0.346%–0.489% interval: the simulation now agrees with the solver within error. A ±0.02-point interval would take roughly **127.3 million rounds**.

REPRODUCE IT

```
node reports/build-blackjack-10m-report.mjs
```

Writes `blackjack-10m.json`, this HTML and the PDF. The seed is 8675309; the 100k and 1M reports are the first 100,000 and 1,000,000 rounds of this sequence.

THE LAST WORD

The interval shrank. The gap disappeared.